NØthing is Logical

Maria Aloni
ILLC & Philosophy
University of Amsterdam
M.D.Aloni@uva.nl

Slides: https://www.marialoni.org/resources/Jinan26.pdf

Shandong University, Jinan

22 October 2026

NØthing is logical (Nihil)

- Goal of the project: a formal account of a class of natural language inferences which deviate from classical logic
- Common assumption: these deviations are not logical mistakes, but consequence of pragmatic enrichments (Grice)
- Initial strategy: develop *logics of conversation* which model next to literal meanings also pragmatic factors and the additional inferences which arise from their interaction
- Novel hypothesis: **neglect-zero** tendency (a cognitive bias rather than a conversational principle) as crucial factor
- Main conclusion: deviations from classical logic consequence of enrichments albeit not (always) of the canonical Gricean kind
  - Dominant view: enrichments always product of **rational interactions** between cooperative language users
  - Nihil hypothesis: enrichments sometimes result from our preference to minimize cognitive effort (**bias-driven** vs product of rational choices)

Nihil team


Anttila


Degano


Klochowicz


Knudstorp


Ramotowska


Yan


Zhou

...

Non-classical inferences

Free choice (FC)

(1) FC: $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond\alpha \wedge \Diamond\beta$ [von Wright 1968]

(2) Deontic FC inference [Kamp 1973]

a. You may go to the beach *or* to the cinema.
b. $\rightsquigarrow$ You may go to the beach *and* you may go to the cinema.

(3) Epistemic FC inference [Zimmermann 2000]

a. Mr. X might be in Victoria *or* in Brixton.
b. $\rightsquigarrow$ Mr. X might be in Victoria *and* he might be in Brixton.

Ignorance

(4) The prize is either in the garden *or* in the attic $\rightsquigarrow$ The speaker doesn't know which [Grice 1989, p.45]

(5) ? I have two *or* three children.

- In the standard approach, ignorance is a conversational implicature
- Less consensus on free choice inferences analysed as conversational implicatures; grammatical (scalar) implicatures; semantic entailments; ...

Paradox of Free Choice

Challenge 1: Explosion

- FC inference invalid in classical modal logic but robust in natural language:

  (6) a. You can have cake or ice cream $\rightsquigarrow$ You can have cake and you can have ice cream

  b. FC: $\Diamond(a \vee b) \rightsquigarrow \Diamond a \wedge \Diamond b$

- Adding FC to classical modal logic (ML) implies the equivalence of any two possibility claims [Kamp 1973]

$$\Diamond a \Rightarrow_{\textit{or}\text{-intro}+\Diamond\text{-mon}} \Diamond(a \vee b) \Rightarrow_{\text{FC}} \Diamond b$$

Challenge 2: Double Prohibition (DP)

- FC effects systematically disappear in negative contexts [Alonso-Ovalle 2006]

  (7) a. You can't have cake or ice cream $\rightsquigarrow$ You can't have cake and you can't have ice cream

  b. DP: $\neg\Diamond(a \vee b) \rightsquigarrow \neg\Diamond a \wedge \neg\Diamond b$

- The combination of FC & DP leads again to explosion [Goldstein 2019]

FC: $\Diamond(a \vee b) \models \Diamond b \Rightarrow_{\textit{contraposition}} \neg\Diamond b \models \neg\Diamond(a \vee b)$

DP: $\neg\Diamond(a \vee b) \models \neg\Diamond a \Rightarrow_{\textit{transitivity}} \neg\Diamond b \models \neg\Diamond a \Rightarrow_{\textit{contraposition}} \Diamond a \models \Diamond b$

Novel hypothesis: neglect-zero

- FC and ignorance inferences are [≠ semantic entailments]
  - Not the result of Gricean reasoning [≠ conversational implicatures]
  - Not the effect of applications of covert grammatical operators [≠ grammatical (scalar) implicatures]
- They are rather a consequence of something else speakers do in conversation, namely,

  NEGLECT-ZERO
  when interpreting a sentence speakers construct models depicting reality (some verifying the sentence, some falsifying it) ↦ common assumption
  and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*) ↦ novel hypothesis

勿空

- Tendency to neglect zero-models follows from the cognitive difficulty of:
  1. conceiving emptiness, the absence of things rather than their presence
  2. evaluating truths with respect to empty witness sets

[Nieder 2016; O. Bott *et al* 2019]

Novel hypothesis: neglect-zero

Illustration

(8) Less than three squares are black.

a. Verifier: [■, □, ■]
b. Falsifier: [■, ■, ■]
c. Zero-models: [□, □, □]; [■, ■, ■]; . . .

Zero-models in (8-c) verify the sentence by virtue of an empty set of black squares

- Cognitive difficulty of zero-models confirmed by experimental findings and connected to / can be argued to explain:
  - the special status of 0 among the natural numbers [Nieder 2016]
  - why downward-monotonic quantifiers are more costly to process than upward-monotonic ones (*less* vs *more*) [O. Bott *et al* 2019]
- NZ hypothesis: neglect-zero also at the origin of many common departures from classical reasoning
  1. FC and ignorance [MA 2022]
  2. Existential Import: every A is B ⇒ some A is B
  3. Principles from connexive logic: [Ziegler, Knudstorp & MA 2026]
     - Aristotle's Thesis: if not A then A ⇒ ⊥
     - Boethius' Thesis: if A then B & if A then not B ⇒ ⊥
  4. ESQ inferences & modified numerals [MA & vOrtmondt 2023]; Non-monotonic attitudes [Yan 2024]; SDA in counterfactuals [Klochowicz 2026], . . .

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(9) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(10) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(11) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(12) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. Zero-models: ?

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(13) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(14) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(15) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(16) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (16-c): verify the sentence by virtue of an empty witness for one of the disjuncts

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(17) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(18) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(19) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(20) M ate an apple or a banana.

a. Verifier: [🍎 | 🍌] ⇐ 'split'

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (20-c): sentence verified by virtue of an empty witness for one of the disjuncts
- **Split state** in (20-a): multiple (possibly conflicting) alternatives processed in a parallel fashion
- Neglect-zero hypothesis: Zero-models, where only one of the disjuncts is depicted, are cognitively taxing and therefore disregarded
- The exclusion of zero-models from the set of verifiers for a disjunction leads directly to an account of ignorance inferences and FC

A new conjecture: no-split

(21) Maria ate an apple or a banana.

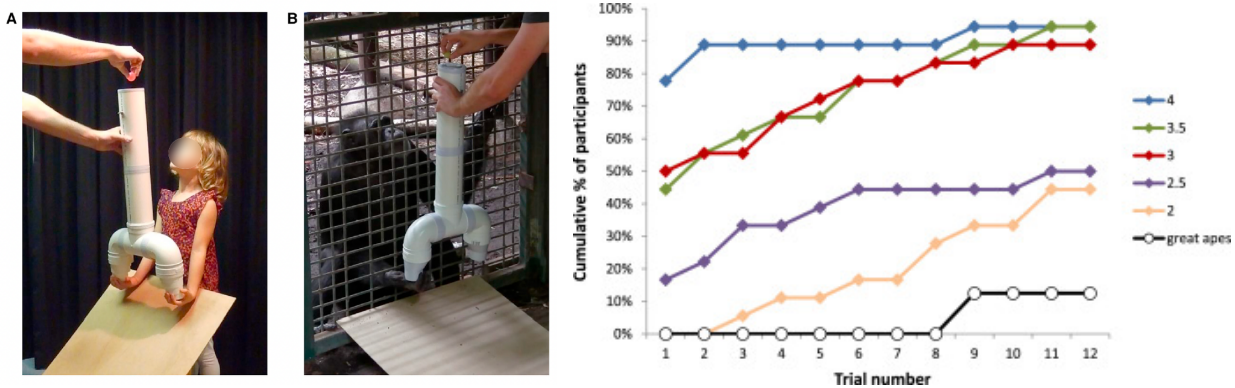
a. Verifier: [🍎 | 🍌] [⇐ **split state**]

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- **Split states**: multiple (possibly conflicting) alternatives processed in a parallel fashion ↦ also a cognitively taxing operation

NO-SPLIT CONJECTURE [Klochowicz, Sbardolini & MA, SuB 2025]
the ability to split states (entertain multiple possibilities) is developed late



Preschoolers have trouble conceiving possible different futures [Redshaw & Suddendorf 2016]

- Proposal: Combination of neglect-zero + no-split can explain conjunctive interpretations of disjunction observed in pre-school children

No-split: conjunctive children

- Children (3-5/6y) sometimes (but systematically) interpret disjunctions conjunctively [Evans 93; Singh *et al* 16 (cf. Skordos *et al* 20); Cochard25; Bleotu *et al* 25]

(22) M ate an apple *or* a banana $\rightsquigarrow$ M ate an apple *and* a banana
$\alpha \vee \beta \rightsquigarrow \alpha \wedge \beta$

(23) M *can* eat an apple *or* a banana $\rightsquigarrow$ M *can* eat an apple *and* a banana
$\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond(\alpha \wedge \beta) \not\equiv \Diamond\alpha \wedge \Diamond\beta$

(24) M did *not* eat apples *or* bananas $\rightsquigarrow$ M *neither* ate apples *nor* bananas
$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$

- Proposal: children have conjunctive readings as they (similarly to adults) neglect zero and, unlike adults, lack the ability to split

1. Deriving ignorance:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{SPLIT}}$ [🍎 | 🍌]

   $\rightsquigarrow$ It might be an apple and it might be a banana (adults)

2. Deriving conjunctive reading:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [🍎 🍌]

   $\rightsquigarrow$ Both an apple and a banana (children)

3. In case of incompatible alternatives: [Leahy & Carey 2020, Bleotu *et al*]

   Left OR right $\Rightarrow_{\text{NZ}}$ ↙ + ↘ $\Rightarrow_{\text{NO-SPLIT}}$ **contradiction** ($\bot$)

   $\rightsquigarrow$ Sequential or first disjunctive interpretations (children)

Cognitive bias approach (NIHIL$^+$)

Common assumption: Reasoning and understanding of natural language involve the creation of mental models [Johnson-Laird 1983, Mascarenhas & Koralus 2013, a.o.]

- **Understanding** a sentence S means being able to mentally construct a model picturing the world which verifies S, and possibly also a model which falsifies it
- **Reasoning** depends on two main processes: first construct verifying models for the premises and then check the validity of the conclusion on these models

Novel hypothesis: biases can constrain the construction of these models and therefore impact both reasoning and interpretation:

- NEGLECT-ZERO prevents the constructions of zero-models;
- NO-SPLIT expresses a dispreference for split-states;
- ...

Cognitive bias approach (NIHIL$^+$)

Comparison with competing accounts

	Ignorance	**FC & DIST**	**ES-Quant**	**Scalar impl.**	**Conj *or* & homog.**
Neo-Gricean	reasoning	reasoning	reasoning	reasoning	—
Grammatical	debated	grammar	grammar	grammar	grammar
NIHIL$^+$	neglect-zero	neglect-zero	neglect-zero	—	negl-z + no-split

- Standard ALT-based derivation of **scalar implicatures** [Grice 75, Horn 72]

  (25) Maria ate an apple or a banana $\rightsquigarrow$ not both
  - a. Literal meaning: $\alpha \vee \beta$ [inclusive]
  - b. ALT: $\alpha \wedge \beta$
  - c. Negation of ALT: $\neg(\alpha \wedge \beta)$ [exclusive]

- NIHIL$^+$: **conversational inferences** sometimes bias-driven (*fast, effortless*) rather than product of alternative-based reasoning (*slow, costly*)

NEXT

- Logical modelling of biases in team semantics
- Experimental findings
  1. Degano *et al* (Nat Lang Sem, 2025): ignorance ⇐
  2. NIHIL *et al* (SuB24-25, CogSci25, XPRAG25): scalar, DIST & ES-Quant
  3. Bleotu *et al* (TbiLLC 2025, SuB25): on conjunctive *or*

Modelling biases in team semantics

General methodology

Natural language sentences translated into classical logic formulas interpreted in a team semantics which models both classical and bias-driven interpretations

NAT LANGUAGE $\mapsto$ LOGICAL LANG

α (**classical**) $\Rightarrow$
$\Rightarrow$ Gric
$\Rightarrow$ ean
$\Rightarrow$ Prag
$[\alpha]^+$ (negl-zero) $\Rightarrow$ mat
$[\alpha]^{++}$ (no-split) $\Rightarrow$ ics
$\ldots$ $\Rightarrow$

Back to FC challenges

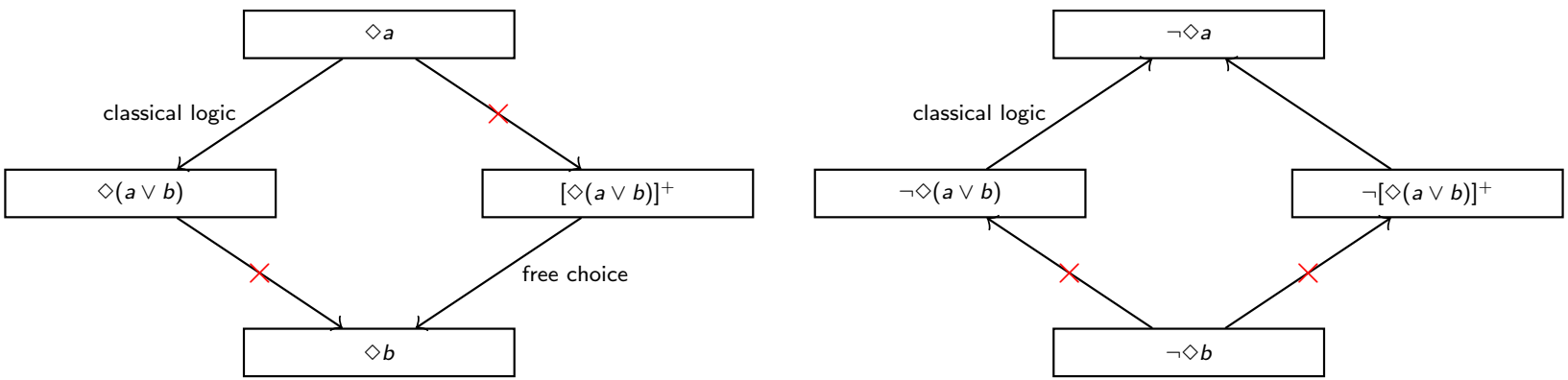


Figure: FC derived only for neglect-zero (NZ) enriched formulas; NZ enrichment vacuous under negation $\mapsto$ failure of contraposition

Modelling biases in team semantics

Team semantics

- Formulas interpreted wrt a set of points of evaluation (a team) rather than single ones [Hodges 1997; Väänänen 2007; Yang & Väänänen 2016]
  - Classical modal logic: $[M = \langle W, R, V \rangle]$

$$M, w \models \phi, \text{ where } w \in W$$

  - Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

- Two crucial features
  - The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ zero-models
  - Multi-membered teams can model parallel processing of alternatives $\mapsto$ split states

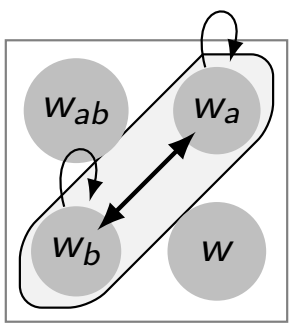


Figure: A multi-membered team modelling the parallel processing of possibilities *a* & *b*

Modelling biases in team semantics

Team semantics

- Formulas interpreted wrt a set of points of evaluation (a team) rather than single ones [Hodges 1997; Väänänen 2007]
  - Classical modal logic: $[M = \langle W, R, V \rangle]$

$$M, w \models \phi, \text{ where } w \in W$$

  - Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

- Two crucial features
  - The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ zero-models
  - Multi-membered teams can model parallel processing of alternatives $\mapsto$ split states

Neglect-zero & no-split

- Model-theoretic method: biases modelled by disallowing empty (neglect-zero) and multi-membered teams (no-split)
- Syntactic method: biases modelled via new logical atoms/operators
  - Neglect-zero: via **non-emptiness atom** NE which disallows empty teams as possible verifiers [Yang & Väänänen 2017]

$$M, t \models \text{NE} \text{ iff } t \neq \emptyset$$

  - No-split: via **flattening operator** F which induces pointwise evaluations and therefore mimics ban on parallel processing of alternatives

$$M, t \models \text{F}\phi \text{ iff for all } w \in t : M, \{w\} \models \phi$$

Modelling biases in team semantics

Classical and enriched interpretations

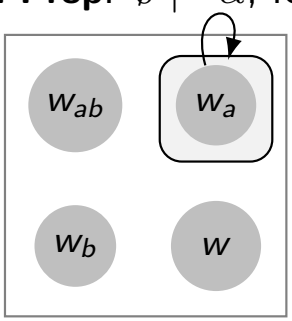
- $\alpha \Rightarrow$ empty team & split allowed $\mapsto$ **classical (literal) meaning**
- $[\alpha]^+ \Rightarrow$ empty team not allowed $\mapsto$ NZ-**enriched**
- $[\alpha]^{++} \Rightarrow$ neither empty team nor split allowed $\mapsto$ NZ+NS-**enriched**

Tensor disjunction in team semantics

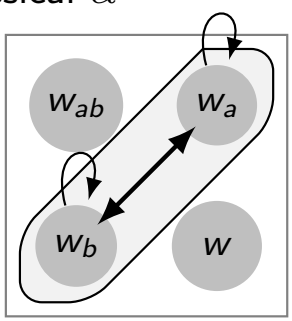
- Team t supports a **disjunction** iff t is the union of two subteams, each supporting one of the disjuncts
  - $a \vee b$: empty team allowed $\mapsto$ subteams can be empty (**classical**)
  - $[a \vee b]^+$: empty team not allowed $\mapsto$ subteams cannot be empty (NZ-**enr**)
  - $[a \vee b]^{++}$: neither empty team nor split allowed $\mapsto$ no empty or distinct subteams allowed (NZ+NS-**enriched**)

Empty Team Prop: $\emptyset \models \alpha$, for all classical α

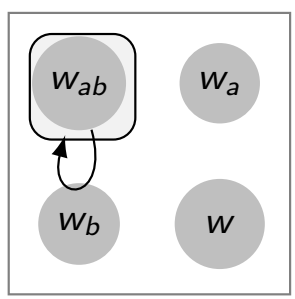
[**PROP**$= \{a, b\}; W = \{w_{ab}, w_a, w_b, w\}$]



$\models a \vee b$
$\not\models [a \vee b]^+$
$\not\models [a \vee b]^{++}$
$\mapsto$ **Zero-model**



$\models a \vee b$
$\models [a \vee b]^+$
$\not\models [a \vee b]^{++}$
$\mapsto$ **No-zero**, **split**



$\models a \vee b$
$\models [a \vee b]^+$
$\models [a \vee b]^{++}$
$\mapsto$ **No-zero**, **no-split**

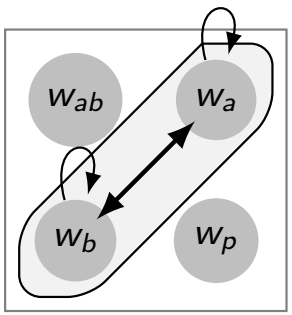
Modelling biases in team semantics: ignorance & conjunctive 'or'

(26) Maria ate an apple or a banana. $[a \vee b]$

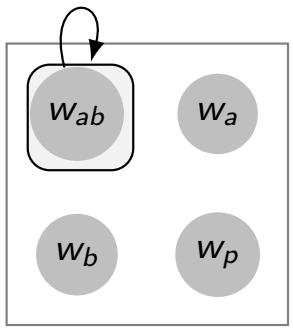
a. **No-zero**, **split** verifier: [🍎 | 🍌] $\rightsquigarrow$ it might be apple, it might be banana

b. **No-zero**, **no-split** verifier: [🍎 🍌] $\rightsquigarrow$ both apple and banana

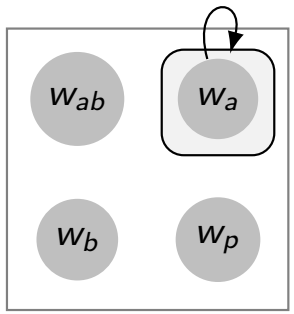
c. **Zero-models**: [🍎] ; [🍌] d. Falsifier: [🍐] $\rightsquigarrow$ neither apple nor banana



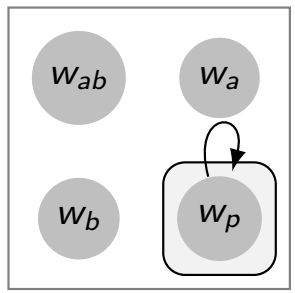
No-zero, **split** verifier



No-zero, **no-split** verifier



Zero-model



Falsifier

Predictions

We derive ignorance & conjunctive 'or' interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(27) M ate an apple or a banana $\rightsquigarrow_{nz}$ It might be an apple and it might be a banana

$[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$ (where R is state-based)

(28) M ate an apple or a banana $\rightsquigarrow_{ns}$ M ate both apple and banana

$[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$

(29) M didn't eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M ate neither apple nor banana

$[\neg(\alpha \vee \beta)]^{(++)} \models \neg\alpha \wedge \neg\beta$

and more predictions including FC, homogeneity, ...

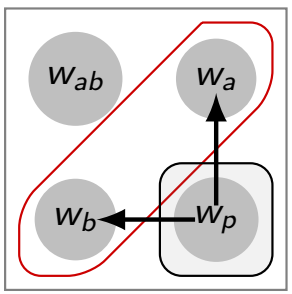
Modelling biases in team semantics: free choice

(30) Maria may eat an apple or a banana. $[\Diamond(a \vee b)]$

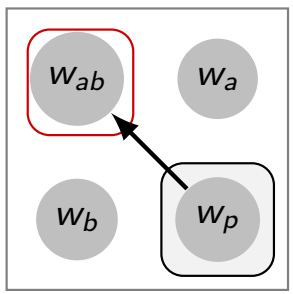
a. **No-zero**, **split** verifier: [🍎 | 🍌] $\rightsquigarrow$ may eat apple, may eat banana

b. **No-zero**, **no-split** verifier: [🍎 🍌] $\rightsquigarrow$ may eat both apple and banana

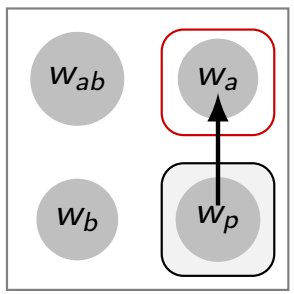
c. **Zero-models**: [🍎] ; [🍌] d. Falsifier: [🍐] $\rightsquigarrow$ may eat neither



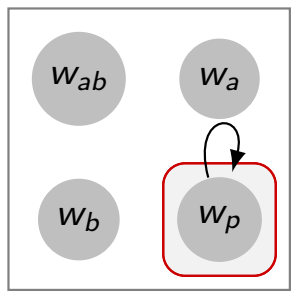
No-zero, **split** verifier



No-zero, **no-split** verifier



Zero-model



Falsifier

Predictions

We derive free choice & conjunctive permission interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(31) M may eat an apple or a banana $\rightsquigarrow_{nz}$ It may be an apple and it may be a banana

$[\Diamond(\alpha \vee \beta)]^+ \models \Diamond\alpha \wedge \Diamond\beta$

(32) M may eat an apple or a banana $\rightsquigarrow_{ns}$ M may eat both apple and banana

$[\Diamond(\alpha \vee \beta)]^{++} \models \Diamond(\alpha \wedge \beta)$

(33) M may not eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M may eat neither apple nor banana

$[\neg\Diamond(\alpha \vee \beta)]^{(++)} \models \neg\Diamond\alpha \wedge \neg\Diamond\beta$

$\Rightarrow$ Freedom of choice requires ability to conceive possible alternative futures

A closer look at these predictions

Many enrichments triggered by disjunction

(34) Maria ate an apple or a banana $(\alpha \vee \beta) \rightsquigarrow$

a. Ignorance: speaker doesn't know which
b. Conjunctive interpretation: both $(\alpha \wedge \beta)$
c. Exclusive interpretation: not both $\neg(\alpha \wedge \beta)$

Can we capture all these enrichments as nz- or ns-effects?

Two components of full ignorance: possibility vs uncertainty

(35) Maria ate an apple or a banana $\rightsquigarrow$ speaker doesn't know which

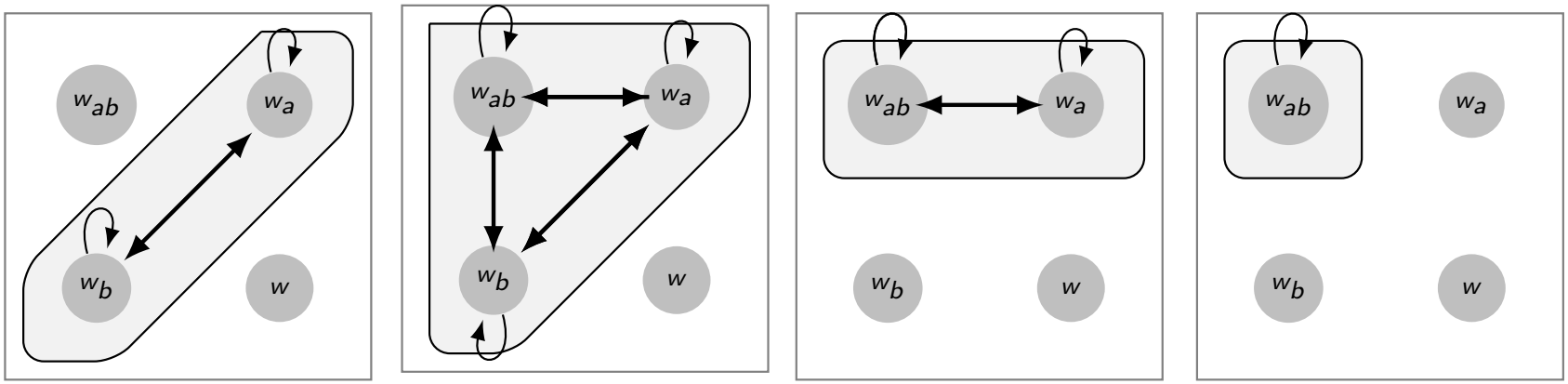
[Degano *et al* 2025][1]

a. Possibility: It is possible that M ate an apple and it is possible that M ate a banana $\Diamond_e\alpha \wedge \Diamond_e\beta$
b. Uncertainty: It is uncertain that M ate an apple and it is uncertain that M ate a banana $\neg\Box_e\alpha \wedge \neg\Box_e\beta$

[1] Degano, Marty, Ramotowska, MA, Sudo, Romoli, Breheny. The ups and downs of ignorance. *Natural Language Semantics* (2025)

Neglect-zero effects on disjunction: predictions of $[\ ]^{+\cdots}$-enrichment

Many no-zero verifiers for nz-enriched disjunction



(a) no-zero & excl $\models \neg(a \wedge b)$ (b) no-z & non-excl $\not\models \neg(a \wedge b)$ (c) no-z & no-uncertain $\not\models \neg\Box_e a$ (d) no-zero & no-split $\models (a \wedge b)$

Figure: Models for enriched $[a \vee b]^+$.

1. Neglect-zero enrichment derives **possibility**: $[\alpha \vee \beta]^+ \models \Diamond_e\alpha \wedge \Diamond_e\beta$
2. Neglect-zero + no-split derives conjunctive *or*: $[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$
3. Neglect-zero does not derive exclusive readings: $[\alpha \vee \beta]^+ \not\models \neg(\alpha \wedge \beta)$
4. Neglect-zero does not derive uncertain inferences: $[\alpha \vee \beta]^+ \not\models \neg\Box_e\alpha$

   $\mapsto$ in contrast to standard neo-Gricean approach to ignorance $\Leftarrow$

Two derivations of full ignorance

1. **Standard neo-Gricean derivation** [Sauerland 2004]

   (i) Uncertainty derived through quantity reasoning

   (36) $\alpha \vee \beta$ ASSERTION

   (37) $\neg\Box_e\alpha \wedge \neg\Box_e\beta$ UNCERTAINTY (from QUANTITY)

   (ii) Possibility derived from uncertainty and quality about assertion

   (38) $\Box_e(\alpha \vee \beta)$ QUALITY ABOUT ASSERTION

   (39) $\Rightarrow \Diamond_e\alpha \wedge \Diamond_e\beta$ POSSIBILITY

2. **Neglect-zero derivation**

   (i) Possibility derived as neglect-zero effect

   (40) $\alpha \vee \beta$ ASSERTION

   (41) $\Diamond_e\alpha \wedge \Diamond_e\beta$ POSSIBILITY (from NEGLECT-ZERO)

   (ii) Uncertainty derived from possibility and scalar reasoning

   (42) $\neg(\alpha \wedge \beta)$ SCALAR IMPLICATURE

   (43) $\Rightarrow \neg\Box_e\alpha \wedge \neg\Box_e\beta$ UNCERTAINTY

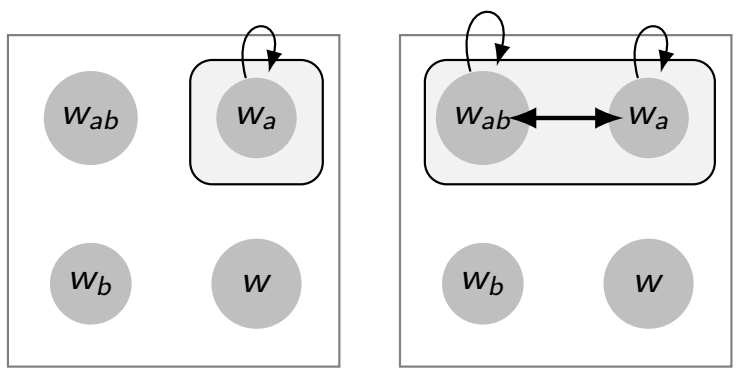
The ups and downs of ignorance [Degano et al 2025]

Contrasting predictions of competing accounts of ignorance

- Neo-Gricean: No possibility without uncertainty
- Neglect-zero: Possibility derived independently from uncertainty

Experimental findings

- Using adapted mystery box paradigm, compared conditions in which
  - both uncertainty and possibility are false [**zero-model**]
  - uncertainty false but possibility true [no-zero, no-uncertain model]
- Zero-models far more rejected (66%) than no-uncertain model (5%)
  $\Rightarrow$ Evidence that possibility can arise without uncertainty
- A challenge for the traditional neo-Gricean approach



(a) **zero-model** (b) no-uncertain

Figure: Models for $(a \vee b)$

Conclusions

- FC, ignorance inference: a mismatch between logic and language
- Grice's insight:
  - stronger meanings can be derived paying more "attention to the nature and importance to the conditions governing conversation"
- Nihil hypothesis: some strengthenings due to cognitive tendencies rather than rational choices
  - FC, possibility and related inferences as neglect-zero effects

    Literal meanings (classical fragment) + cognitive factor (NZ) $\Rightarrow$ FC, possibility, connexive principles, etc

  - Conjunctive *or*, homogeneity as no-zero + no-split effect

    Literal meanings (classical fragment) + cognitive factors (NZ, NS) $\Rightarrow$ conjunctive *or*, homogeneity

- Implementation in team-based modal logics
- Ups and downs of ignorance: a challenge for the neo-Gricean approach

THANK YOU![2]

[2]This work is supported by NWO OC project *Nothing is Logical* (grant no 406.21.CTW.023).

Appendix

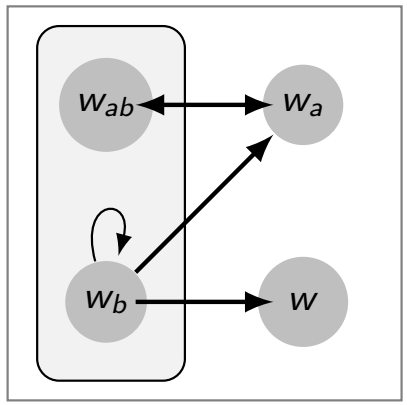
BSML: Classical Modal Logic + NE

Language

$$\phi := p \mid \neg\phi \mid \phi \vee \phi \mid \phi \wedge \phi \mid \Diamond\phi \mid \text{NE}$$

Bilateral team semantics

Given Kripke model $M = \langle W, R, V\rangle$ & teams/states $s, t, t' \subseteq W$



$$\begin{array}{rll}
M, s \models p & \text{iff} & \text{for all } w \in s : V(w, p) = 1 \\
M, s =\!| p & \text{iff} & \text{for all } w \in s : V(w, p) = 0 \\
M, s \models \neg\phi & \text{iff} & M, s =\!| \phi \\
M, s =\!| \neg\phi & \text{iff} & M, s \models \phi \\
M, s \models \phi \vee \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \;\&\; M, t \models \phi \;\&\; M, t' \models \psi \qquad \Leftarrow \\
M, s =\!| \phi \vee \psi & \text{iff} & M, s =\!| \phi \;\&\; M, s =\!| \psi \\
M, s \models \phi \wedge \psi & \text{iff} & M, s \models \phi \;\&\; M, s \models \psi \\
M, s =\!| \phi \wedge \psi & \text{iff} & \text{there are } t, t' : t \cup t' = s \;\&\; M, t =\!| \phi \;\&\; M, t' =\!| \psi \\
M, s \models \Diamond\phi & \text{iff} & \text{for all } w \in s : \text{there is } t \subseteq R[w] : t \neq \emptyset \;\&\; M, t \models \phi \\
M, s =\!| \Diamond\phi & \text{iff} & \text{for all } w \in s : M, R[w] =\!| \phi \qquad [\text{where } R[w] = \{v \in W \mid wRv\}] \\
M, s \models \text{NE} & \text{iff} & s \neq \emptyset \\
M, s =\!| \text{NE} & \text{iff} & s = \emptyset
\end{array}$$

Illustrations $\{w_{ab}, w_b\} \not\models a$; $\not\models \neg a$, but $\models a \vee \neg a$; $\{w_{ab}, w_b\} \models b$; but $\not\models \Diamond b$; $\models \Diamond a$;
$\emptyset \models a$; $\emptyset \models b$; $\emptyset =\!| a$; $\emptyset =\!| b$; $\emptyset \models \Diamond a$; $\emptyset \models \Diamond b$; $\emptyset =\!| \Diamond a$; $\emptyset =\!| \Diamond b$; $\emptyset \not\models \text{NE}$

Fact (**Empty team property**). Every NE-free (classical) formula α in BSML is supported & anti-supported by the empty team: $M, \emptyset \models \alpha$ & $M, \emptyset =\!| \alpha$

BSML: some results

Proof Theory: BSML [MA, Anttila & Yang, 2024]
`https://doi.org/10.1215/00294527-2024-0027`

Expressive completeness: BSML exp. complete for the class of all convex union-closed modal properties invariant under bounded bisimulation [Anttila & Knudstorp 25]
`https://arxiv.org/abs/2503.21850`

Comparisons via translation into MIL: BSL$_\blacklozenge$ [Knudstorp *et al* 2026]
`https://arxiv.org/abs/2610.15680`

Proof Theory: BSL$^*_\blacklozenge$ [Menorello 2026]
`https://www.illc.uva.nl/Research/Publications/Reports/MoL/`



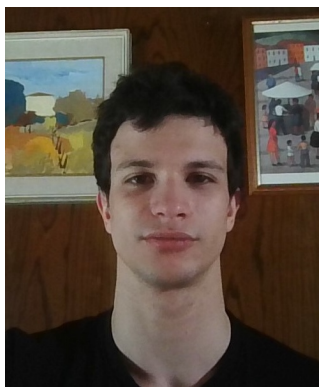
Aleksi Anttila



Søren Knudstorp



Fan Yang



E. Menorello

Implementations: two neglect-zero enrichment functions

- Two ways to model neglect-zero in BSML: [MA 2022]
  1. $[\,]^+$: enriches formulas with the requirement to satisfy NE distributed along each of their subformulas $\mapsto$ BSML$^+$
  2. $[\,]^*$: equivalent to ruling out $\emptyset$ from the set of possible teams $\mapsto$ BSML*

- For NE-free α,
  1. $[\alpha]^+$ defined as follows: [MA 2022]

$$\begin{aligned}
[p]^+ &= p \wedge \mathrm{NE} \\
[\neg\alpha]^+ &= \neg[\alpha]^+ \wedge \mathrm{NE} \\
[\alpha \vee \beta]^+ &= ([\alpha]^+ \vee [\beta]^+) \wedge \mathrm{NE} \\
[\alpha \wedge \beta]^+ &= ([\alpha]^+ \wedge [\beta]^+) \wedge \mathrm{NE} \\
[\Diamond\alpha]^+ &= \Diamond[\alpha]^+ \wedge \mathrm{NE}
\end{aligned}$$

  2. $[\alpha]^*$ defined as follows: [Klochowicz 2026]

$$\begin{aligned}
[p]^* &= (p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^* &= \neg([\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^* &= (([\alpha]^* \vee [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^* &= (([\alpha]^* \wedge [\beta]^*) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^* &= (\Diamond[\alpha]^* \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

[*-implementation $\equiv$ ruling out $\emptyset$ from the set of possible teams]

Implementations: two neglect-zero enrichment functions

- Both implementations derive possibility & free choice inferences:
  - Possibility: $[\alpha \vee \beta]^{+/*} \models \Diamond_e \alpha \wedge \Diamond_e \beta$ [R is state-based]
  - Free choice: $[\Diamond(\alpha \vee \beta)]^{+/*} \models \Diamond\alpha \wedge \Diamond\beta$
- But also strong 'neither' readings under negation:
  - Not or: $[\neg(\alpha \vee \beta)]^{+/*} \models \neg\alpha \wedge \neg\beta$
  - Double Prohibition: $[\neg\Diamond(\alpha \vee \beta)]^{+/*} \models \neg\Diamond\alpha \wedge \neg\Diamond\beta$
    [Challenge 2: failure of contraposition]
- But still differences under negation:

$\neg[p]^{+} = \neg(p \wedge \mathrm{NE}) = \neg p \vee \neg\mathrm{NE} = \neg p \vee \bot = \neg p$

$\neg[p]^{*} = \neg((p \wedge \mathrm{NE}) \vee \neg\mathrm{NE}) = \neg(p \wedge \mathrm{NE}) \wedge \neg\neg\mathrm{NE} = (\neg p \vee \neg\mathrm{NE}) \wedge \mathrm{NE} = \neg p \wedge \mathrm{NE}$

- Only BSML* derives negative possibility and negative free choice:
  - Negative possibility: $[\neg(\alpha \wedge \beta)]^{*} \models \Diamond_e\neg\alpha \wedge \Diamond_e\neg\beta$ [R is state-based]
  - Negative FC: $[\neg\Box(\alpha \wedge \beta)]^{*} \models \Diamond\neg\alpha \wedge \Diamond\neg\beta$

Implementations: adding no-split

- Implementation of no-split: uses flattening operator F

$$M, s \models \mathrm{F}\phi \text{ iff for all } w \in s : M, \{w\} \models \phi$$

$$M, s \mathrel{=\!\!|} \mathrm{F}\phi \text{ iff for all } w \in s : M, \{w\} \mathrel{=\!\!|} \phi$$

  Flattening $\mapsto$ formulas always interpreted wrt to singleton states

- Facts:
  1. OR $\Rightarrow$ AND: $\mathrm{F}[\alpha \vee \beta]^{+/*} \models \alpha \wedge \beta$
  2. Conjunctive FC: $\Diamond\mathrm{F}[\alpha \vee \beta]^{+/*} \models \Diamond(\alpha \wedge \beta)$ $\quad [\not\models \Box(\alpha \wedge \beta)]$
  3. NOT OR: $\neg\mathrm{F}[\alpha \vee \beta]^{+/*} \models \neg\alpha \wedge \neg\beta$
  4. NOT AND: $\neg\mathrm{F}[\alpha \wedge \beta]^{*} \models \neg\alpha \wedge \neg\beta$, but $\neg\mathrm{F}[\alpha \wedge \beta]^{+} \not\models \neg\alpha \wedge \neg\beta$
- Both BSML$^+$ and BSML* derive:
  - $\mapsto$ conjunctive *or* in combination with flattening (1-2)
  - $\mapsto$ classical disjunctive interpretation under negation (3)
- But only BSML* predicts
  - 'Neither' readings for NOT AND in combination with F (4) [Sbardolini 2023]
- Only in BSML$^+$, $\emptyset$ is needed as part of the building blocks

  $\Rightarrow$ Natural to assume BSML* for "conjunctive" children who plausibly do not yet have full access to $\emptyset$

Implementations: no-split enrichment

For NE- and F-free α, $[\alpha]^{**}$ defined as follows:

$$\begin{aligned}
[p]^{**} &= (\mathrm{F}p \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\neg\alpha]^{**} &= (\mathrm{F}\neg[\alpha]^{**} \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \vee \beta]^{**} &= (\mathrm{F}([\alpha]^{**} \vee [\beta]^{**}) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\alpha \wedge \beta]^{**} &= (\mathrm{F}([\alpha]^{**} \wedge [\beta]^{**}) \wedge \mathrm{NE}) \vee \neg\mathrm{NE} \\
[\Diamond\alpha]^{**} &= (\mathrm{F}\Diamond[\alpha]^{**} \wedge \mathrm{NE}) \vee \neg\mathrm{NE}
\end{aligned}$$

Predictions

1. OR $\Rightarrow$ AND: $[\alpha \vee \beta]^{**} \models \alpha \wedge \beta$
2. Conjunctive FC: $\Diamond[\alpha \vee \beta]^{**} \models \Diamond(\alpha \wedge \beta)$ $\quad [\not\models \Box(\alpha \wedge \beta)]$
3. NOT OR: $\neg[\alpha \vee \beta]^{**} \models \neg\alpha \wedge \neg\beta$
4. NOT AND: $\neg[\alpha \wedge \beta]^{**} \models \neg\alpha \wedge \neg\beta$

BSML & related systems: info states (= teams) vs possible worlds

- Failure of bivalence in BSML

$$M, s \not\models p \ \& \ M, s \not\models \neg p, \text{ for some info state } s$$

- Info states: less determinate than possible worlds
  - just like truthmakers, situations, possibilities, HYPE states, ...
- Technically:
  - Truthmakers/possibilities/HYPE states: points in a partially ordered set
  - Info states: sets of possible worlds, also elements of a partially ordered set, the Boolean lattice $Pow(W)$
- Thus systems using these structures are closely connected, although might diverge in motivation:
  - HYPE, truthmaker & possibility semantics: description of ontological structures in the world
  - BSML & inquisitive semantics: explaining patterns in inferential & communicative human activities
- NEXT:
  - Comparison via translations in Modal Information Logic [vBenthem19]

BSML & related systems: comparisons via translation

- Modal Information Logic (MIL) (van Benthem, 1989, 2019):[3] common ground where related systems can be interpreted and their connections and differences can be explored
- Goal: translations into (extensions of) MIL of the following systems:
  - Possibility semantics (Humberstone, Holliday)
  - Truthmaker semantics (Fine)
  - BSML
  - Inquisitive semantics (Ciardelli, Groenendijk & Roelofsen)
  - HYPE (Leitgeb)

  (cf. Gödel's (1933) translation of intuitionistic logic into modal logic)
- Here focus on propositional fragments:
  - disjunction
  - negation
- New result: translation of BSL (propositional fragment of BSML) in an extension of MIL

Søren Knudstorp, Maria Aloni, Alexandru Baltag, Johan van Benthem, Nick Bezhanishvili (2026). *Teams, Truthmakers, Translations*. `https://arxiv.org/abs/2610.15680`

[3] van Benthem, Implicit and Explicit Stances in Logic, *Journal of Philosophical Logic* (2019)

Modal Information Logic (MIL)

Language

$$\phi \quad ::= \quad p \mid \neg\phi \mid \phi \wedge \phi \mid \phi \vee \phi \mid \langle sup \rangle \phi\psi$$

where $p \in \mathsf{P}_\pm := \{p_+, p_- \mid p \in \mathsf{P}\}$.

Models and interpretation

Formulas are interpreted on triples $M = (X, \leq, V)$ where $\leq$ is a partial order

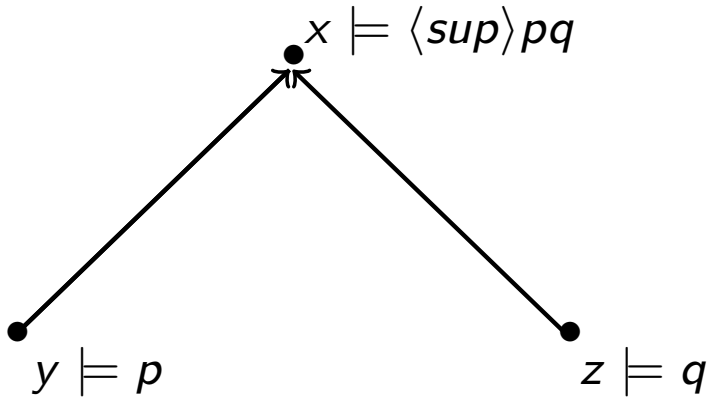
$$\begin{array}{rll}
\mathcal{M}, x \models p & \text{iff} & x \in V(p) \\
\mathcal{M}, x \models \neg\phi & \text{iff} & \mathcal{M}, x \not\models \phi \\
\mathcal{M}, x \models \phi \wedge \psi & \text{iff} & \mathcal{M}, x \models \phi \text{ and } \mathcal{M}, x \models \psi \\
\mathcal{M}, x \models \phi \vee \psi & \text{iff} & \mathcal{M}, x \models \phi \text{ or } \mathcal{M}, x \models \psi \\
\mathcal{M}, x \models \langle sup \rangle \phi\psi & \text{iff} & \text{there are } y, z : x = sup_\leq(y, z) \ \&\ \mathcal{M}, y \models \phi \ \&\ \mathcal{M}, z \models \psi
\end{array}$$

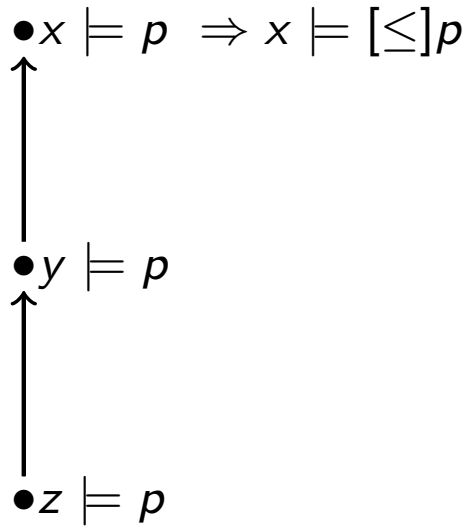
$$[\leq]\phi = \neg\langle sup \rangle(\neg\varphi)\top$$

$$\begin{array}{rll}
\mathcal{M}, x \models [\leq]\phi & \text{iff} & \text{for all } y : y \leq x \ \Rightarrow\ \mathcal{M}, y \models \phi
\end{array}$$

Modal Information Logic (MIL)

Examples





Translations into Modal Information Logic

- Possibility semantics (Humberstone, Holliday)[4]

$$\begin{aligned} tr(p) &= [\leq]\langle\leq\rangle p \\ tr(\neg\phi) &= [\leq]\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= [\leq]\langle\leq\rangle(tr(\phi) \vee tr(\psi)) \\ &\ldots \end{aligned}$$

- Truthmaker semantics (Fine)[5] (MIL on (distributive) semilattices)

$$\begin{aligned} (p)^+ &= p_+ \\ (p)^- &= p_- \\ (\neg\phi)^+ &= (\phi)^- \\ (\neg\phi)^- &= (\phi)^+ \\ (\phi \wedge \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\ (\phi \wedge \psi)^- &= (\phi)^- \vee (\psi)^- \\ (\phi \vee \psi)^+ &= (\phi)^+ \vee (\psi)^+ \\ (\phi \vee \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^- \end{aligned}$$

[4] van Benthem, Bezhanishvili, Holliday, A bimodal perspective on possibility semantics. *Journal of Logic and Computation* (2017)

[5] van Benthem, Implicit and Explicit Stances in Logic. *Journal of Philosophical Logic* (2019)

Translations into (extended) Modal Information Logic

- Truthmaker semantics (Fine) (MIL on (distributive) semilattices)

$$
\begin{aligned}
(p)^+ &= p_+ \\
(p)^- &= p_- \\
(\neg\phi)^+ &= (\phi)^- \\
(\neg\phi)^- &= (\phi)^+ \\
(\phi \wedge \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\
(\phi \wedge \psi)^- &= (\phi)^- \vee (\psi)^- \\
(\phi \vee \psi)^+ &= (\phi)^+ \vee (\psi)^+ \\
(\phi \vee \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^-
\end{aligned}
$$

- Classical fragment BSL (MIL* on distributive semilattices)

$$
\begin{aligned}
(p)^+ &= [\leq]\langle \mathrm{sup}^*\rangle p_+ \\
(p)^- &= [\leq]\langle \mathrm{sup}^*\rangle p_- \\
(\neg\phi)^+ &= (\phi)^- \\
(\neg\phi)^- &= (\phi)^+ \\
(\phi \wedge \psi)^+ &= (\phi)^+ \wedge (\psi)^+ \\
(\phi \wedge \psi)^- &= \langle sup\rangle(\phi)^-(\psi)^- \\
(\phi \vee \psi)^+ &= \langle sup\rangle(\phi)^+(\psi)^+ \\
(\phi \vee \psi)^- &= (\phi)^- \wedge (\psi)^-
\end{aligned}
$$

$\mathcal{M}, x \models \langle \mathrm{sup}^*\rangle\phi$ iff there exist $x_1, \ldots, x_n$ s.t. each $x_i \models \varphi$ & $x = sup_{\leq}\{x_1 \cdots x_n\}$

Translations into (extended) Modal Information Logic

- Inquisitive semantics (Groenendijk, Roelofsen & Ciardelli)

$$\begin{aligned} tr(p) &= [\leq]\langle \mathrm{sup}^* \rangle p \\ tr(\neg\phi) &= [\leq]\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= tr(\phi) \vee tr(\psi) \\ tr(\phi \to \psi) &= [\leq](tr(\phi) \to tr(\psi)) \end{aligned}$$

- HYPE (Leitgeb)[6] (bimodal perspective needed for negation)

$$\begin{aligned} tr(p) &= [\leq]p \\ tr(\neg\phi) &= \blacksquare\neg tr(\phi) \\ tr(\phi \wedge \psi) &= tr(\phi) \wedge tr(\psi) \\ tr(\phi \vee \psi) &= tr(\phi) \vee tr(\psi) \\ tr(\phi \to \psi) &= [\leq](tr(\phi) \to tr(\psi)) \end{aligned}$$

$\mathcal{M}, x \models \blacksquare\phi$ iff for all y: $xCy \Rightarrow \mathcal{M}, y \models \varphi$

Given a compatibility model $\mathcal{M} = (X, C, \leq, V)$

[6] Odintsov, S., Wansing, H. Routley Star and Hyperintensionality. J Philos Logic (2021)

Disjunction and Negation

- Three notions of disjunction expressible in (extended) MIL:
  - Boolean disjunction: $\phi \vee \psi$
    [classical logic, intuitionistic logic, inquisitive logic]
  - Lifted/tensor/split disjunction: $\langle sup \rangle \phi\psi$
    [BSML, dependence logic, team semantics, operational semantics for Positive R]
  - Cofinal disjunction: $[co](\phi \vee \psi)$ (where $[co]\phi =: [\leq]\langle \leq \rangle \phi$)
    [possibility semantics, dynamic semantics]
- Four notions of negation:
  - Boolean negation: $\neg\phi$
    [classical logic, ...]
  - Bilateral negation: $(\neg\phi)^+ = (\phi)^- \ \& \ (\neg\phi)^- = (\phi)^+$
    [truthmaker semantics, BSML, ...]
  - Intuitionistic-like negation: $[\leq]\neg\phi$
    [possibility semantics, inquisitive semantics, intuitionistic logic]
  - Modal negation: $\blacksquare\neg\phi$
    [HYPE, ...]
- Some combinations:
  - Boolean disjunction + boolean negation $\mapsto$ classical logic
  - Boolean notions in other combinations can generate non-classicality:
    - Boolean disjunction + intuitionistic negation $\mapsto$ intuitionistic/inquisitive logic[7]
  - Classicality also generated by non-boolean combinations:
    - Split disjunction + bilateral negation (classical fragm. BSPL)
    - Cofinal disjunction and intuitionistic negation (possibility semantics)

[7] Inquisitive & intuitionistic logic: same connectives but different translations for the atoms.

Full translation for BSPL

$$
\begin{array}{rcl@{\qquad}rcl}
p^+ & := & [\leq]\langle \mathsf{sup}^* \rangle p_+ & p^- & := & [\leq]\langle \mathsf{sup}^* \rangle p_- \\
\mathrm{NE}^+ & := & \bigwedge_{p \in \mathrm{P}} \neg(p^+ \wedge p^-) & \mathrm{NE}^- & := & \bigwedge_{p \in \mathrm{P}} (p^+ \wedge p^-) \\
\bot^+ & := & \mathrm{NE}^- & \bot^- & := & \top \\
(\neg\varphi)^+ & := & \varphi^- & (\neg\varphi)^- & := & \varphi^+ \\
(\varphi \vee \psi)^+ & := & \langle \mathsf{sup} \rangle \varphi^+ \psi^+ & (\varphi \vee \psi)^- & := & \varphi^- \wedge \psi^- \\
(\varphi \wedge \psi)^+ & := & \varphi^+ \wedge \psi^+ & (\varphi \wedge \psi)^- & := & \langle \mathsf{sup} \rangle \varphi^- \psi^-
\end{array}
$$

Theorem (Knudstorp 2026)

$$\varphi \models_{BSPL} \psi \qquad \text{iff} \qquad \Gamma, \varphi^+ \models_{MIL^*} \psi^+$$

where

$$\Gamma := \left\{ [\leq](\mathrm{NE}^+ \vee \mathrm{NE}^-), \bigwedge_{p \in \mathrm{P}} \langle \mathsf{sup} \rangle p^+ p^- \right\}$$

Selected References

Aloni, Maria (2022). "Logic and conversation: The case of free choice". In: Semantics and Pragmatics 15.5, pp. 1–60. DOI: 10.3765/sp.15.5.

Aloni, Maria, Aleksi Anttila, and Fan Yang (2024). "State-based Modal Logics for Free Choice". In: Notre Dame Journal of Formal Logic 65.4, pp. 367–413. DOI: https://doi.org/10.1215/00294527-2024-0027.

Aloni, Maria and Peter van Ormondt (2023). "Modified numerals and split disjunction: the first-order case". In: Journal of Logic, Language and Information 32.4, pp. 539–567. DOI: 10.1007/s10849-023-09399-w.

Benthem, Johan van (2019). "Implicit and Explicit Stances in Logic". In: Journal of Philosophical Logic 48.3, pp. 571–601. DOI: 10.1007/s10992-018-9485-y. URL: https://doi.org/10.1007/s10992-018-9485-y.

Benthem, Johan van, Nick Bezhanishvili, and Wesley H. Holliday (Aug. 2016). "A bimodal perspective on possibility semantics". In: Journal of Logic and Computation 27.5, pp. 1353–1389. ISSN: 0955-792X. DOI: 10.1093/logcom/exw024. eprint: https://academic.oup.com/logcom/article-pdf/27/5/1353/18492005/exw024.pdf. URL: https://doi.org/10.1093/logcom/exw024.

Bott, Oliver, Fabian Schlotterbeck, and Udo Klein (2019). "Empty-Set Effects in Quantifier Interpretation". In: Journal of Semantics 36 (1), pp. 99–163.

Ciardelli, Ivano, Jeroen Groenendijk, and Floris Roelofsen (2018). Inquisitive Semantics. Oxford University Press.

Degano, Marco et al. (2025). "The ups and downs of ignorance". In: Natural Language Semantics 33, pp. 1–41. DOI: 10.1007/s11050-024-09226-3.

Fine, Kit (2017). "Truthmaker Semantics". In: A Companion to the Philosophy of Language. Ed. by B. Hale, C. Wright, and A. Miller. Wiley, pp. 556–77.

Goldstein, Simon (Nov. 2019). "Free choice and homogeneity". In: Semantics and Pragmatics 12.23, pp. 1–53. DOI: 10.3765/sp.12.23.

Selected References

Grice, Herbert Paul (1975). "Logic and Conversation". In: Syntax and Semantics, Volume 3: Speech Acts. Ed. by Peter Cole and Jerry Morgan. Academic Pr, pp. 41–58.

— (1989). Studies in the Way of Words. Harvard University Press.

Humberstone, I.L. (1981). "From worlds to possibilities". In: Journal of Philosophical Logic 10.3, pp. 313–399.

Johnson-Laird, Philip N. (1983). Mental Models. Cambridge University Press.

Kamp, Hans (1973). "Free Choice Permission". In: Proceedings of the Aristotelian Society 74, pp. 57–74.

Klochowicz, Tomasz, Giorgio Sbardolini, and Maria Aloni (2026). "Cognitive bias approach to the acquisition of disjunction". In: Proceedings of Sinn und Bedeutung. Vol. 30, pp. 554–571. DOI: https://doi.org/10.18148/sub/2026.v30.1353.

Klochowicz, Tomasz et al. (2025). "Neglect zero: evidence from priming across constructions". In: Proceedings of CogSci 2025. URL: https://escholarship.org/uc/item/36w6x7z9.

Koralus, Philipp and Salvador Mascarenhas (2013). "The Erotetic theory of reasoning: bridges between formal semantics and the psychology of deductive inference". In: Philosophical Perspectives 27.1, pp. 312–365. DOI: https://doi.org/10.1111/phpe.12029.

Leahy, Brian P. and Susan E. Carey (2020). "The Acquisition of Modal Concepts". In: Trends in Cognitive Sciences 24.1, pp. 65–78. DOI: https://doi.org/10.1016/j.tics.2019.11.004.

Leitgeb, Hannes (2019). "HYPE: A System of Hyperintensional Logic (with an Application to Semantic Paradoxes)". In: Journal of Philosophical Logic 2, pp. 305–405.

Nieder, Andreas (2016). "Representing Something Out of Nothing: The Dawning of Zero". In: Trends in Cognitive Sciences 20 (11), pp. 830–842.

Redshaw, Jonathan and Thomas Suddendorf (2016). "Children's and Apes' Preparatory Responses to Two Mutually Exclusive Possibilities". In: Current Biology 26.13, pp. 1758–1762. DOI: https://doi.org/10.1016/j.cub.2016.04.062.

Wright, G.H. von (1968). An Essay on Deontic Logic and the Theory of Action. North Holland.

Selected References

Zimmermann, Ede (2000). "Free Choice Disjunction and Epistemic Possibility". In: Natural Language Semantics 8, pp. 255–290.